

A2 consolidation – global well-posedness of the scalar Brazovskii gradient flow

TECT verification-first repository · claim A2-PDE-WELLPOSED

first issued 2026-06-23 · this version issued 2026-06-23 · v1.1

1. Purpose and scope

This is the single-place consolidation of A2 (the analytical foundation that the TECT free-energy minimisation has a well-posed dynamical realisation), and the entry note of the A2 reproduction bundle. The full proof is [claims/A2-PDE-WELLPOSED/notes/a2-local-global-wellposedness-260623-260623-v1.2.tex.txt](#); the numerics are [codes/foundations/a2_wellposedness_checks.py](#) (8/8 PASS). Scope (operator-approved): ONE real scalar field on a fixed periodic cell $\mathbb{T}^3$, $\mu^2 > 0$, $\gamma > 0$, $\frac{3}{2} < s \leq 2$. NOT full Class-II; NOT the $\mu^2 < 0$ condensate branch; $\mu^2 > 0$ is the disordered-side positive linear gap, not BCC condensation.

2. The theorem

Theorem (A2). *Assume $Y > 0$, $\mu^2 > 0$ and $\gamma > 0$. Then the L^2 gradient flow $\partial_t \phi = -K(-i\nabla)\phi - \lambda\phi^3 - \gamma\phi^5$ on $\mathbb{T}^3$ with $K(q) = \mu^2 + Y(q^2 - q_0^2)^2$ (A1-KERNEL-CONV) has, for every $\phi_0 \in H^s$ with $\frac{3}{2} < s \leq 2$, a unique global solution $\phi \in C([0, \infty); H^s) \cap C^\infty((0, \infty) \times \mathbb{T}^3)$, depending continuously on ϕ_0 ; $F_{\text{TECT}}[\phi(t)]$ is non-increasing for $t > 0$ (for $t \geq 0$ if additionally $\phi_0 \in H^2$).*

3. Proof structure (full proof: v1.2)

(i) $L = K(-i\nabla)$ is sectorial: $\lambda_0 := \min_k K(k) \geq \mu^2 > 0$, so $-L$ generates an analytic semigroup with $X^\beta = H^{4\beta}$ and smoothing $\|L^\beta e^{-tL}\| \leq C_\beta t^{-\beta}$. (ii) $N(\phi) = -\lambda\phi^3 - \gamma\phi^5$ is locally Lipschitz $X^\beta \rightarrow L^2$ for $\beta \in (\frac{3}{8}, \frac{1}{2}]$ ($H^{4\beta} \hookrightarrow L^\infty$, $d = 3$). (iii) Henry's theorem gives local existence/uniqueness + parabolic smoothing. (iv) For global existence, energy is taken from $t_0 > 0$ (parabolic smoothing $\Rightarrow F[\phi(t_0)] < \infty$), $F[\phi(t)] \leq F[\phi(t_0)]$ on $[t_0, T_{\max})$, sextic coercivity ($\gamma > 0$, Young $C_* = C_*(\lambda, \gamma) < \infty$; production anchor 1.01×10^{-2}) $\Rightarrow$ uniform H^2 bound; $H^2 \subseteq H^{4\beta} = X^\beta$ ($\beta \leq \frac{1}{2}$) $\Rightarrow T_{\max} = \infty$.

4. Devil's-advocate (closed in v1.2 review)

(α) $\inf \text{spec } L = \mu^2$ false on a generic cell — replaced by the lower bound $\lambda_0 \geq \mu^2 > 0$. (β) Sobolev direction — the correct $H^2 \subseteq H^{4\beta}$ is used. (γ) range — restricted to $\frac{3}{2} < s \leq 2$ ($s > 2$ needs separate higher-order persistence, not claimed). (δ) energy domain — $F[\phi_0]$ may be undefined for $s < 2$, so the a priori bound is taken from $t_0 > 0$. Quantitative sanity check (production anchor): the constants ($C_* = 1.01 \times 10^{-2}$ etc.) are reproduced by the 8/8 verification; the theorem itself uses only their finiteness.

5. Result footer

Result ID: A2-PDE-WELLPOSED (consolidation)

Precise statement: $Y>0, \mu^2>0, \gamma>0 \Rightarrow$ the scalar Brazovskii L2 gradient flow on T^3 is globally well-posed for ϕ_0 in H^s , $3/2<s\leq 2$: unique global solution, smooth for $t>0$, continuous dependence, F non-increasing for $t>0$.

Scope: scalar, fixed periodic cell, $\mu^2>0, \gamma>0, 3/2<s\leq 2$. NOT Class-II; NOT $\mu^2<0$.

Dependencies: A1-KERNEL-CONV.

Hypotheses: (H0) $Y>0$; (H1) A2-H1-KERNEL-POSITIVITY ($\mu^2>0$); (H2) A2-H2-SEXTIC-COERCIVITY ($\gamma>0$).

Evidence grade: PROVED CONDITIONAL (analytic) + EXECUTED (8/8).

Reproduction command: python codes/foundations/a2_wellposedness_checks.py

Expected output: 8/8 PASS; exit 0.

Tier: T6 PROVED CONDITIONAL (operator-approved 2026-06-23).

Full proof: a2-local-global-wellposedness-260623-260623-v1.2.

No-overclaim statement: No $s>2$ persistence; no Class-II; no $\mu^2<0$; $\mu^2>0$ is NOT BCC condensation; no unique global minimiser.

Next required action: (bundle) referee reproduction; Class-II / $\mu^2<0$ are separate targets.